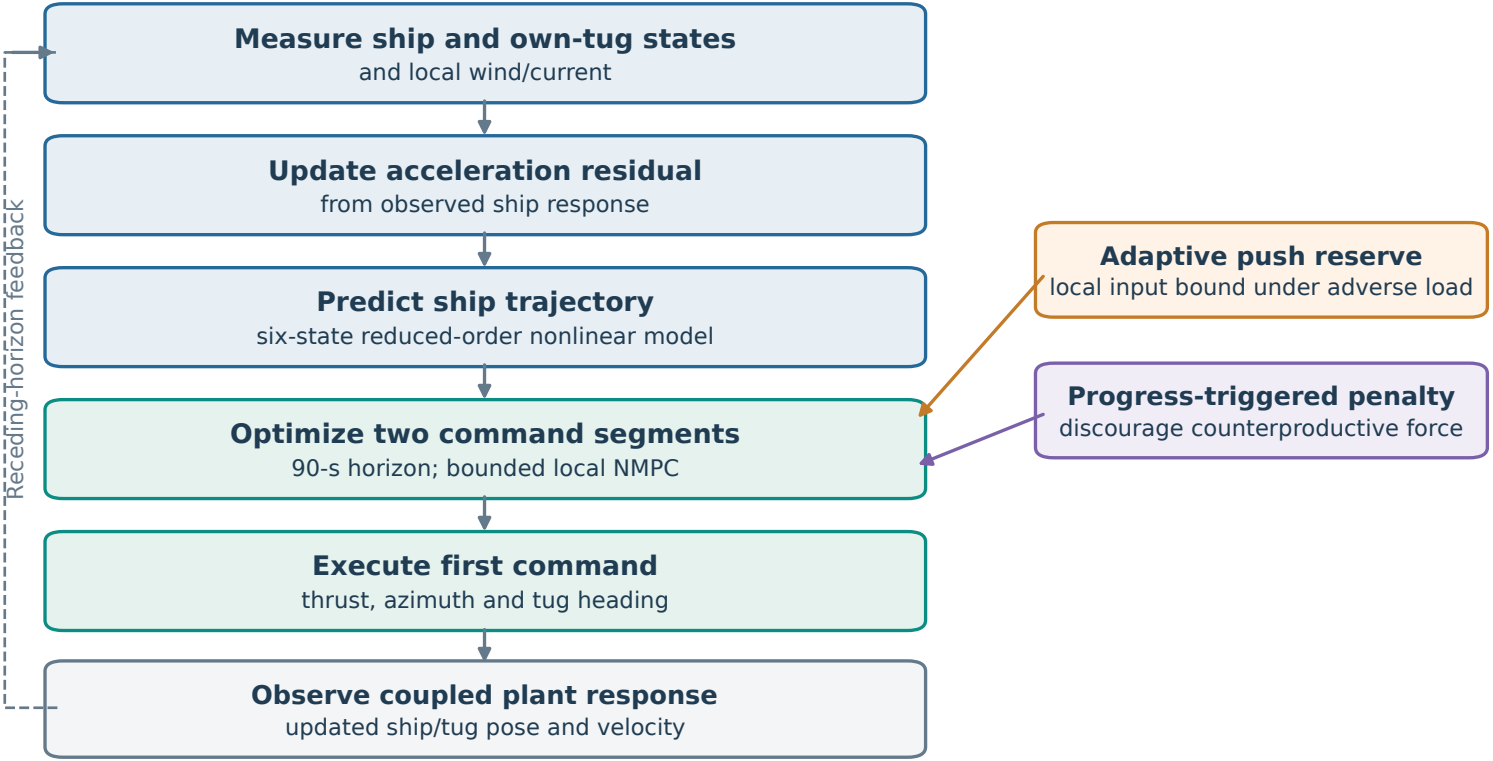
Measure ship and own-tug states
and local wind/current
Update acceleration residual
from observed ship response
Predict ship trajectory
six-state reduced-order nonlinear model
Optimize two command segments
90-s horizon; bounded local NMPC
Execute first command
thrust, azimuth and tug heading
Observe coupled plant response
updated ship/tug pose and velocity
Receding-horizon feedback
Adaptive push reserve
local input bound under adverse load
Progress-triggered penalty
discourage counterproductive force